

Introduction of Engineering Design

Playgrounds are vital for children’s physical and social development, and while current accessibility standards address many needs, we focus on enhancing the experience for children who are blind or visually impaired through specialized tactile maps. These maps allow users to understand environments through touch by representing spatial features with raised surfaces and textures. By using tactile maps, children gain the confidence and autonomy to explore playgrounds independently.



Figure 1. Aerial footage of design playground, captured using drone.



Figure 2. Completed tactile map of design playground.

We developed a standardized protocol for creating 3D printed tactile maps of playgrounds using photogrammetry. This process involves utilizing collected 2D images to obtain spatial information, measurements, and accurate 3D models of a desired area. By improving efficiency and repeatability, we aim to make tactile playground maps more accessible, produceable, and implementable.

Our design goals are to reduce production time by at least 75% while maintaining a dimensional accuracy of at least 90% for the tactile maps relative to the physical playgrounds.

Engineering Design Process Overview

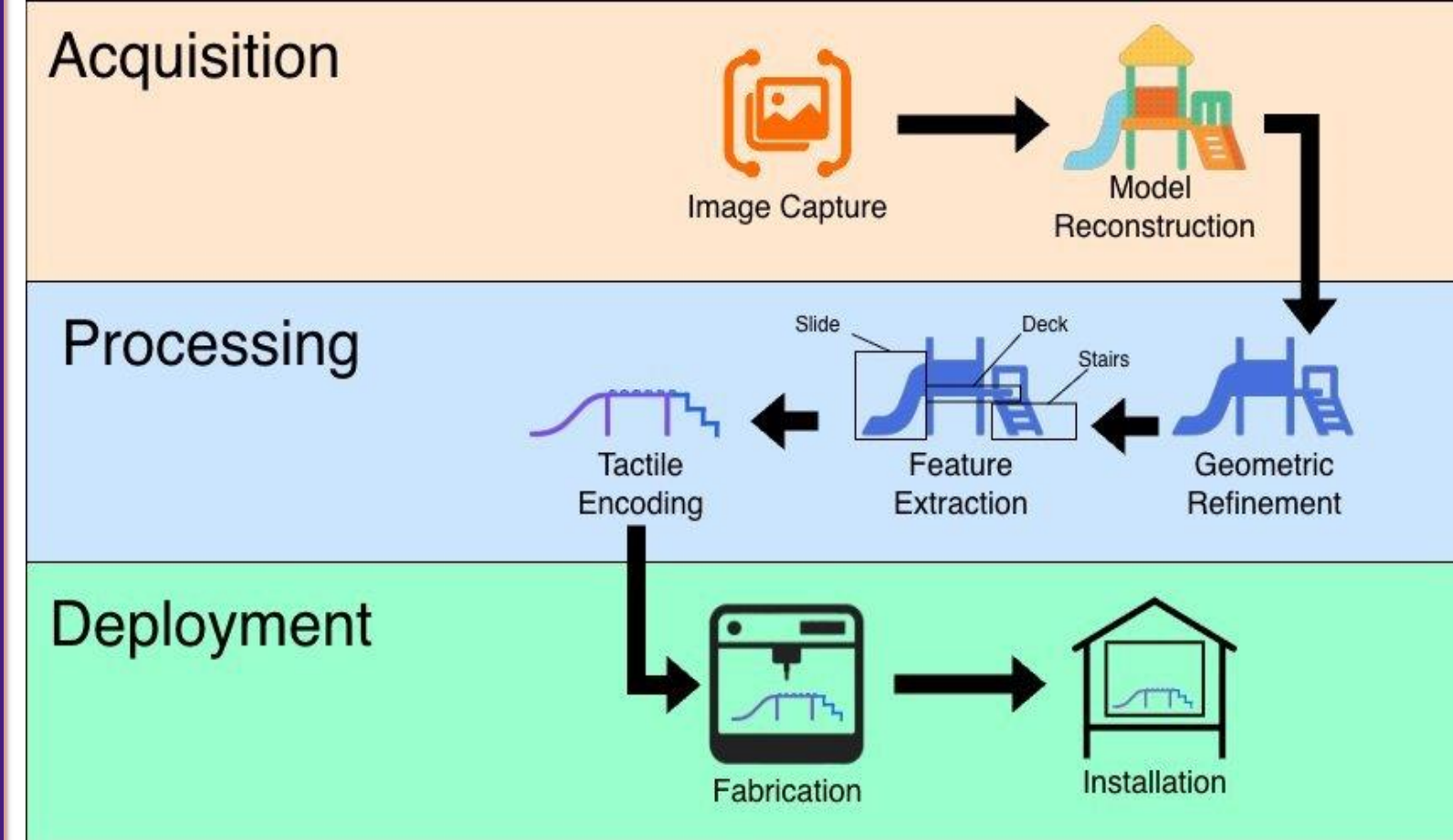


Figure 3. Engineering process diagram for creation of a playground tactile map

Acquisition: A drone and smartphone capture hundreds of overlapping photos of the playground, which are uploaded into 3DF Zephyr to automatically generate a 3D model.
Processing: The model goes through several steps to refine features. The individual components are identified and separated, and tactile textures, Braille labels, and a 40:1 scale are applied to prepare it for printing.
Deployment: The model is 3D printed to be installed at the Louisiana School for the Visually Impaired (LSVI).

Acquisition

Image Capture

Playground data was collected utilizing a DJI Mini 4 Pro drone and iPhone 17 Pro camera systems. 750 –800 high-resolution images and video frames were captured from multiple angles with 70–80% overlap to ensure accurate reconstruction. Elevation changes between datasets also contributed to overall clarity and accuracy. Various camera points that indicate where images were taken were visible in the completed reconstruction.



Figure 4. Drone used for image capture, in flight.

Model Reconstruction

These datasets were processed in the photogrammetry software, 3DF Zephyr, to generate a set of detailed point clouds and meshes. The software utilized numerous filtering protocols that prioritize photo-consistency. This output becomes the backbone of the overall model itself and structures the tactile map.

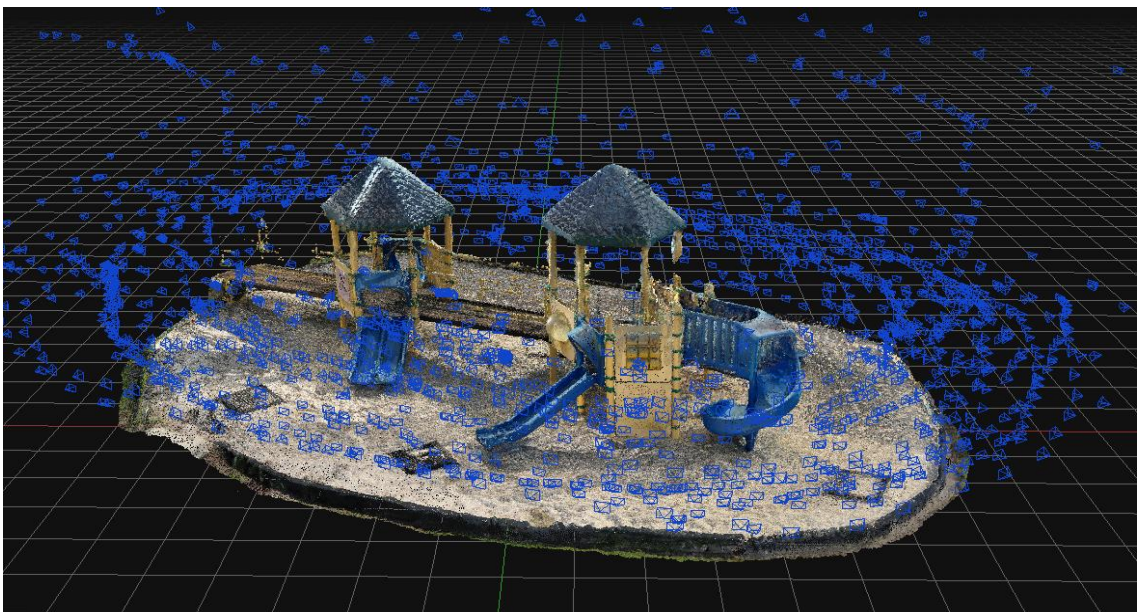


Figure 5. 3-D Playground model reconstruction from captured images in 3DF Zephyr.

Processing

Geometric Refinement

The reconstructed photogrammetry model is run through denoising algorithms to remove scan noise while preserving surfaces. Holes within the mesh are filled to improve surface integrity. Then the repaired mesh is run through surface-preserving global smoothing to remove scan artifacts from surfaces.



Figure 6. 3D model of playground after smoothing operations.

Feature Extraction

The refined model is segmented into functional subcomponents (e.g., ground planes, decks, ramps, and slides) using geometric properties. Planar boundaries are simplified to preserve shape while removing edge irregularities introduced during scanning. Slide regions are identified and bounded for use in tactile model reconstruction.

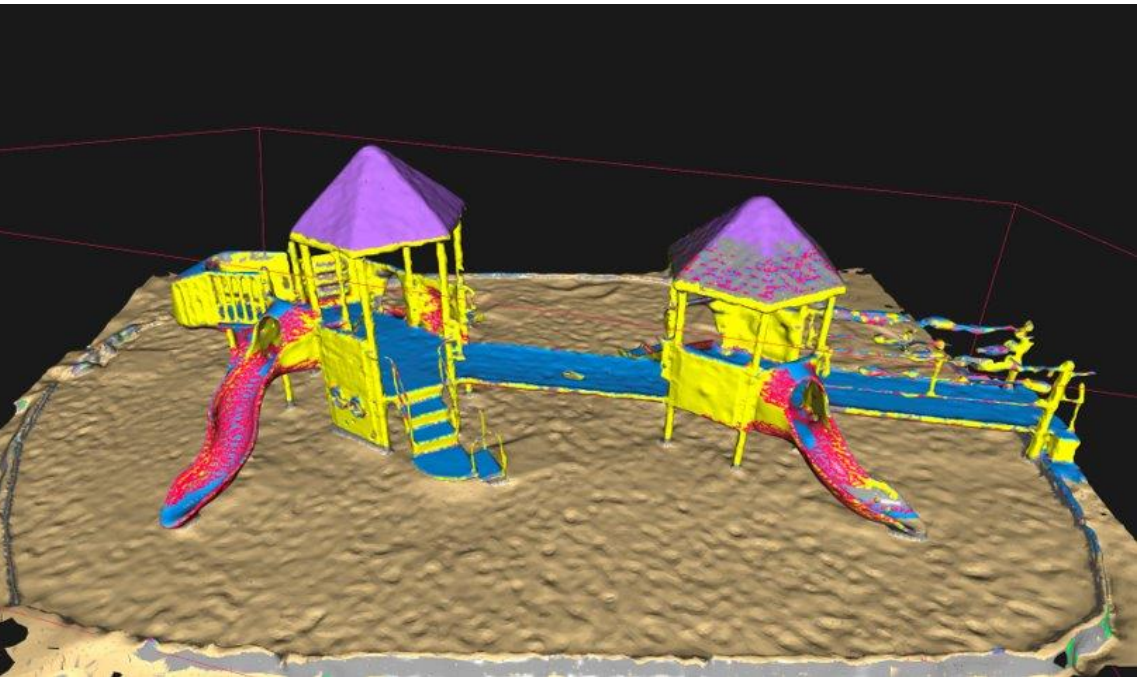


Figure 7. 3-D model of playground with color-coded features and smoothed surfaces.

Tactile Encoding

Platforms are normalized by elevation and extruded to the ground plane. Slide regions are simplified and positioned against platforms. Textures created by the LSU Community Playground Project are applied to planar elements programmatically according to detected and manually-specified classifications.

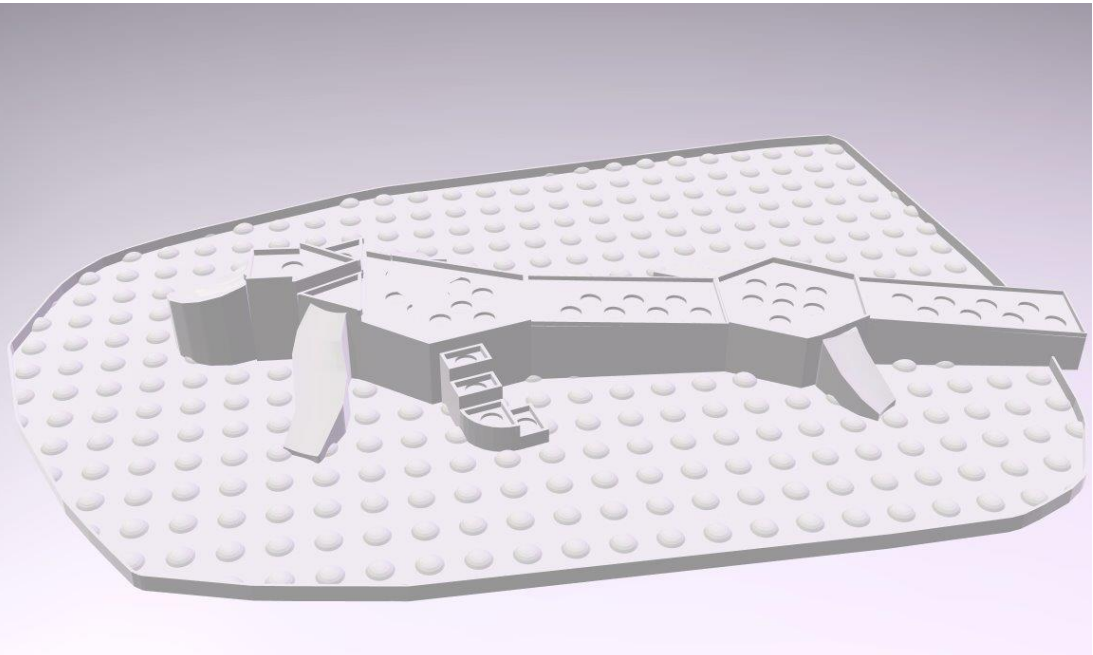


Figure 8. Tactile map with non-essential features removed for simplified use.

Deployment

The physical deployment of the model translates environmental data into a tactile coding system. The finalized model was scaled (1:40) and prepared for fabrication using 3D printing. Prototypes were fabricated using PLA for rapid iteration and verification of tactile textures, while final iterations will implement a more long-term option, providing impact resistance necessary for public use longevity.

The deployment stage prioritizes the implementation of high-contrast surface textures and gradients to clearly define change in material and elevation. The tactile map is designed to improve navigation, safety, and independent play for visually impaired users.

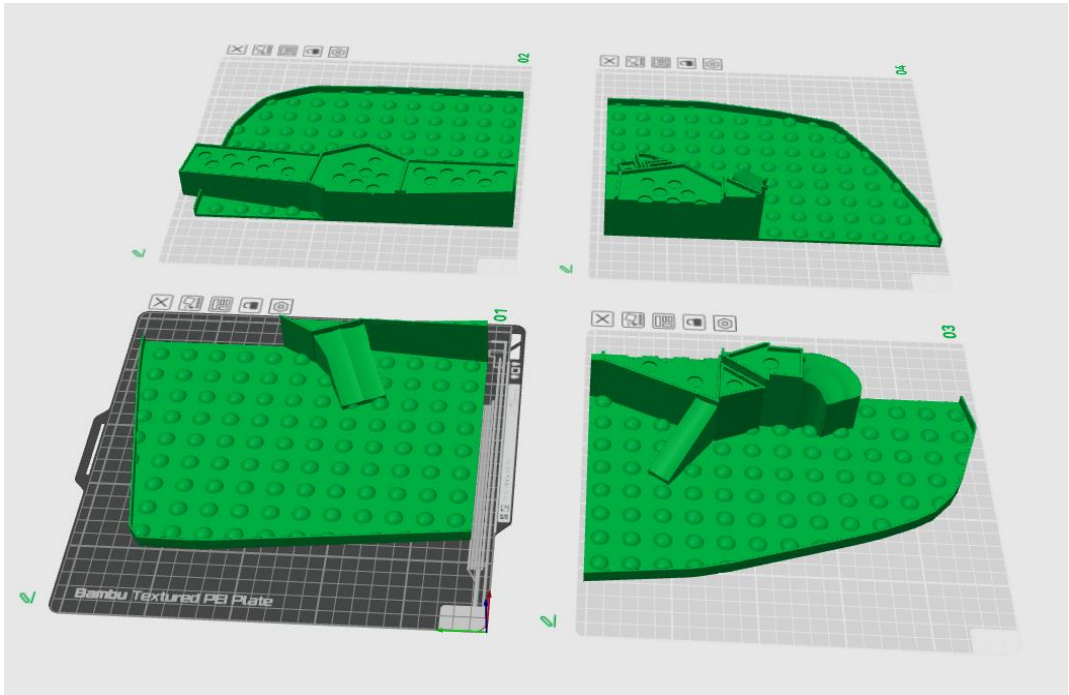


Figure 9. Tactile map is split into sections to comply with 3-D print bed size.



Figure 10. Completed, printed tactile map that is ready for installation and use.

Testing Results

Time Study

To evaluate efficiency, we compared manual measuring and CAD modeling of two LSVI playgrounds against our automated protocol. Our process yielded significant time savings: 81% for the larger playground (reduced from 11 to 2 hours) and 78.9% for the smaller site (reduced from 9 to 2 hours). Notably, because the Zephyr and Processing stages are fully automated, they require minimal active user input, allowing for simultaneous multitasking.

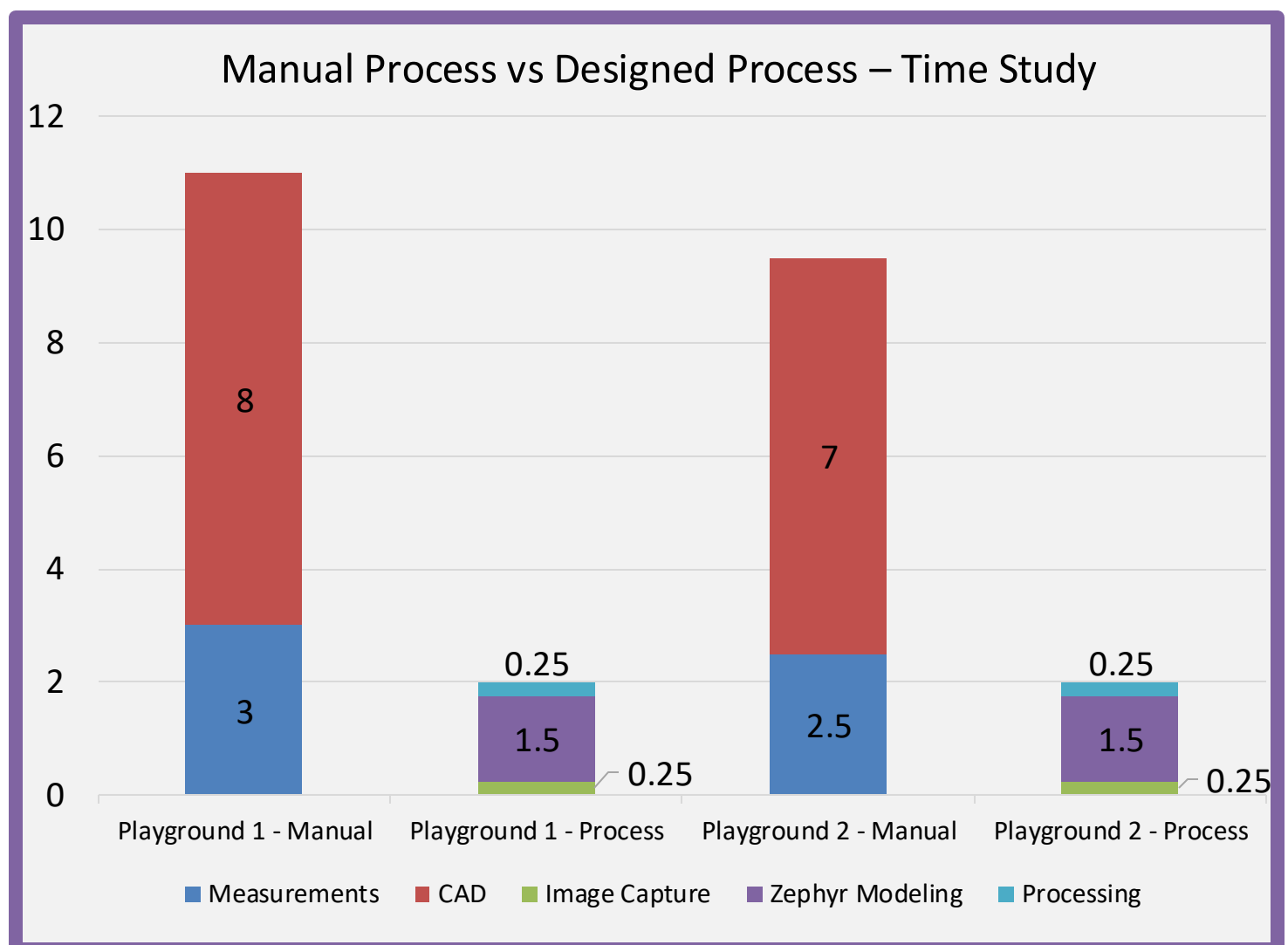


Figure 11. Time study on manual process vs. designed process to show new protocol efficiency.

Percent Accuracy

		Hand Modeled (mm^3)	Overlap (mm^3)	% Accuracy
With Slides	Playground 1	1.907E10	1.904E9	90.0
	Playground 2	1.005E10	1.148E9	88.5
Without Slides	Playground 1	1.453E10	1.301E9	91.0
	Playground 2	6.924E9	6.198E8	91.0

Figure 12. Volumetric accuracy comparison between photogrammetric reconstruction and hand-modeled baseline data for Playground 1 and 2 under "With Slides" and "Without Slides" testing conditions.

Dimensional accuracy was evaluated by overlaying photogrammetric models with hand-modeled counterparts. Accuracy was defined as the ratio of the subtracted volumetric difference to the total model volume. We applied this metric to two playground sites, comparing the results of our automated protocol against hand-modeled data derived from physical measurements. The larger Playground 1 achieved 90% (with slides) and 91% (without), while the smaller Playground 2 achieved 88.5% and 91%. The complex geometry of the slides introduced higher variability and lower accuracy compared to standard playground features.

Conclusions

Our team successfully established a scalable protocol for developing tactile playground maps, designed for broad usability across varying budgets and resource constraints. Testing was conducted to compare the efficiency of the old, manual modeling steps to the new protocol we created. We achieved an 80% reduction in production time, exceeding our initial efficiency target of 75%. Additionally, we found that our process confirmed an average 89.25% accuracy rate with slides and a 91% accuracy rate without slides, successfully meeting our benchmark of 90% dimensional precision.

Acknowledgements

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